30 KW BIDIRECTIONAL SIC ON-BOARD CHARGER

SYS.3 System Architectural Design & Detailed HW/SW Functional Requirements

Rev B0 - Detailed Block Package

Requirement authority: OBC30-VOL1-MRD-001 Rev A0 + OBC-VOL2 A1-A4 / consolidated program documentation | Freeze/authority: Source Parts I/II revision authority; calendar freeze not explicit

Document Field	Controlled Value
Document ID	OBC30-SYS3-ARCH-REQ-001
Revision	Rev B0 - Detailed Block Package
Technical information freeze / authority	Source Parts I/II revision authority; calendar freeze not explicit
Primary source baseline	OBC30-VOL1-MRD-001 Rev A0 + OBC-VOL2 A1-A4 / consolidated program documentation
Process scope	Automotive SPICE 4.0 SYS.3 with downstream HWE.1 / SWE.1 / SYS.4 / SYS.5 / VAL.1 handoff
Architecture blocks	18
Detailed SYS.3 requirements	90
Claim boundary	Derived SYS.3 engineering work product; released OEM requirements, HARA/TSC, homologation and market/grid requirements take precedence.

1. Purpose, Scope and Requirement Authority

This package converts the controlled product baseline into a block-by-block SYS.3 architectural design with explicit hardware/software allocation, algorithms, safety mechanisms, interfaces, verification scenarios and traceability handoff.

The document and companion workbook are intended to be baselined together. A populated trace link or planned test is not executed evidence; closure requires an approved procedure, controlled test configuration, raw result, pass/fail decision and problem/deviation disposition.

ASIL and cybersecurity classifications remain inherited from the approved vehicle/item HARA, FSC/TSC and TARA. This architecture does not create a final ASIL or homologation claim.

2. Frozen / Controlled Technical Baseline

- 30 kW continuous on approved three-phase AC; single-phase operation is derated.
- Bidirectional grid-to-battery and battery-to-grid/load capability where vehicle configuration and regulation enable it.
- Modular 2 x 15 kW architecture.
- Three-level T-type or ANPC bidirectional AFE baseline; two-level AFE remains a comparator branch.
- Two isolated bidirectional DAB or CLLC DC/DC modules.
- Battery concept envelope 250-920 V supporting 400 V and 800 V variants; internal DC-link concept 700-880 V.
- Peak efficiency target >=97.5%; PF >=0.99 and current THD <5% at rated reference condition.
- Vehicle liquid cooling; 15-year/300,000 km design-life target; <=15 kg and <=12 L program targets.

3. High-Level System Architecture

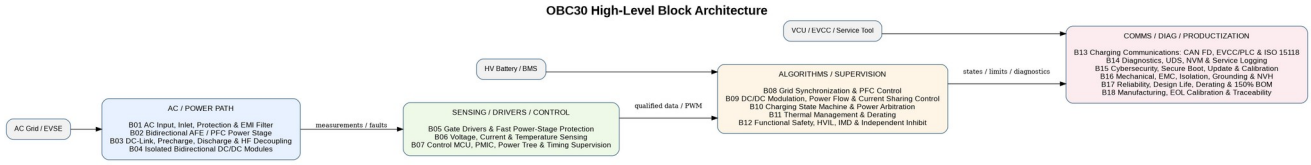


Figure 1 - High-Level Block Architecture

4. End-to-End Algorithm / Information Flow



Figure 2 - Control / Algorithm Information Flow

5. Operating States and Safety Flow

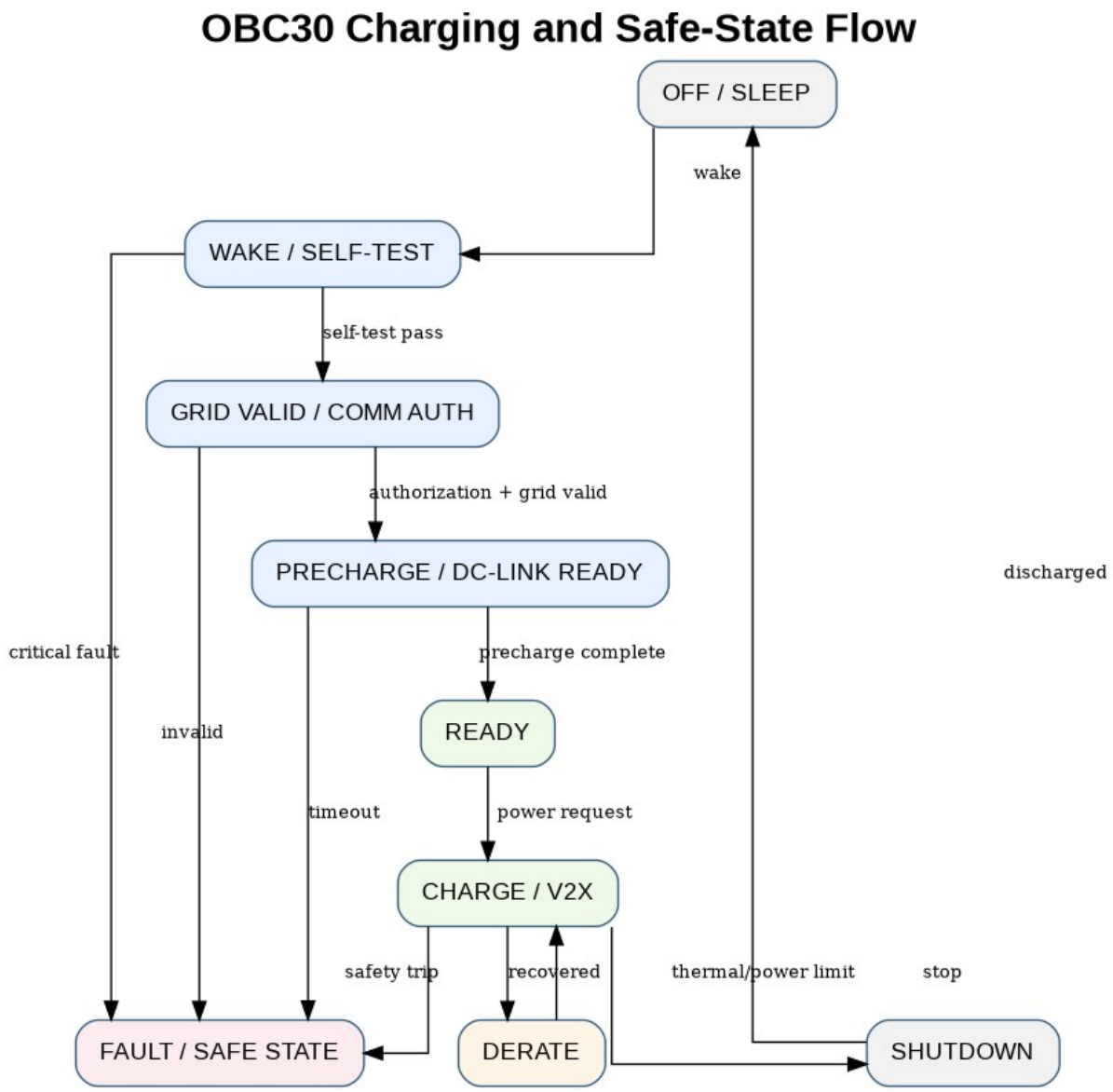


Figure 3 - Operating / Safe-State Flow

6. B01 - AC Input, Inlet, Protection & EMI Filter

AC interface, surge/inrush protection, phase monitoring and conducted/radiated EMI control.

Architecture and boundaries

- Three-phase input capability: The OBC AC interface shall support 30 kW continuous grid-to-battery operation on the released three-phase supply; 400 Vac line-to-line is the reference nominal operating point unless the vehicle ICD releases another value. Acceptance / boundary: 30 kW continuous at released three-phase nominal; exact regional voltage/frequency range controlled by vehicle/market ICD.

- Single-phase derating: The AC input path and controller shall identify single-phase operation and limit current/power according to inlet, cable, EVSE, regional and thermal constraints. Acceptance / boundary: No 30 kW single-phase claim; current remains within released inlet/cable/EVSE limits.
- Input inrush/precharge: The AC/DC input path shall control inrush/precharge such that contactors, rectifier devices, EMI components and DC-link capacitors remain within pulse-current and I2t limits. Acceptance / boundary: No uncontrolled contactor closure into an uncharged DC link; repetitive retry thermal limits enforced.

Algorithm / dynamic behavior

Grid interface qualifier: detect supply -> validate phase/voltage/frequency -> close input path -> monitor phase loss/unbalance -> publish grid-valid state.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B01-01	Three-phase input capability	The OBC AC interface shall support 30 kW continuous grid-to-battery operation on the released three-phase supply; 400 Vac line-to-line is the reference nominal operating point unless the vehicle ICD releases another value.	30 kW continuous at released three-phase nominal; exact regional voltage/frequency range controlled by vehicle/market ICD.	HW/SW	Source-derived	MR-PWR-001
OBC30-SYS3-B01-02	Single-phase derating	The AC input path and controller shall identify single-phase operation and limit current/power according to inlet, cable, EVSE, regional and thermal constraints.	No 30 kW single-phase claim; current remains within released inlet/cable/EVSE limits.	HW/SW	Source-derived	MR-PWR-003
OBC30-SYS3-B01-03	Input inrush/precharge	The AC/DC input path shall control inrush/precharge such that contactors, rectifier devices, EMI components and DC-link capacitors remain within pulse-current and I2t limits.	No uncontrolled contactor closure into an uncharged DC link; repetitive retry thermal limits enforced.	HW/SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B01-04	Phase fault detection	The grid interface shall detect phase loss, severe phase unbalance and out-of-range frequency/voltage before or during power conversion.	Invalid supply inhibits or derates power without unstable current control.	HW/SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B01-05	EMI filter stability	The EMI filter shall meet conducted/radiated emission goals without creating an unstable interaction with the AFE control loop or representative grid/EVSE	Impedance/control stability review closed; no oscillation in released switching modes.	HW	Engineering-derived	MR-EMC-001

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		impedance.				

Safety, diagnostics and failure containment

- Three-phase input capability: 30 kW continuous at released three-phase nominal; exact regional voltage/frequency range controlled by vehicle/market ICD.
- Input inrush/precharge: No uncontrolled contactor closure into an uncharged DC link; repetitive retry thermal limits enforced.
- Phase fault detection: Invalid supply inhibits or derates power without unstable current control.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-01 - STDES-PFCBIDIR 15 kW three-phase three-level bidirectional AFE (STMicroelectronics): source link
- REF-OBC-02 - TIDA-01606 bidirectional three-phase three-level T-type AFE (Texas Instruments): source link

7. B02 - Bidirectional AFE / PFC Power Stage

Two-channel 15 kW-class SiC AFE/PFC implementation for 30 kW three-phase charging and bidirectional power flow.

Architecture and boundaries

- Modular AFE architecture: The 30 kW AFE shall be partitioned as two coordinated 15 kW-class channels or an equivalent released architecture. Acceptance / boundary: Full-power operation maintains defined current sharing; one-channel degraded strategy is explicitly defined.
- Three-level AFE baseline: The preferred bidirectional AFE shall use a three-level T-type or ANPC SiC topology; a two-level solution is an alternate branch requiring efficiency/EMC/thermal revalidation. Acceptance / boundary: Topology selection documented by loss, magnetics, EMC, control, safety and BOM trade study.
- Power factor: The AFE shall control grid current to achieve PF >=0.99 at rated released three-phase charging conditions. Acceptance / boundary: PF >=0.99 at rated condition; light-load behavior characterized.

Algorithm / dynamic behavior

VOC/dq PFC: PLL angle -> abc/dq transform -> DC-link/power reference -> Id/Iq current control -> modulation -> neutral-point balance -> PWM.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B02-01	Modular AFE architecture	The 30 kW AFE shall be partitioned as two coordinated 15 kW-class	Full-power operation maintains defined current sharing;	HW/SW	Source-derived	MR-PWR-001

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		channels or an equivalent released architecture.	one-channel degraded strategy is explicitly defined.			
OBC30-SYS3-B02-02	Three-level AFE baseline	The preferred bidirectional AFE shall use a three-level T-type or ANPC SiC topology; a two-level solution is an alternate branch requiring efficiency/EMC/thermal revalidation.	Topology selection documented by loss, magnetics, EMC, control, safety and BOM trade study.	HW	Source-derived	MR-EFF-001
OBC30-SYS3-B02-03	Power factor	The AFE shall control grid current to achieve PF >=0.99 at rated released three-phase charging conditions.	PF >=0.99 at rated condition; light-load behavior characterized.	HW/SW	Source-derived	MR-PWR-001
OBC30-SYS3-B02-04	Input current THD	The AFE current-control and filter design shall achieve total input-current THD <5% at the defined rated operating point.	THD <5% at rated point; harmonic spectrum recorded across voltage/load corners.	HW/SW	Source-derived	MR-PWR-001
OBC30-SYS3-B02-05	Bidirectional AFE power flow	Where the bidirectional variant is released, the AFE shall support controlled DC-to-AC power flow under valid grid and authorization conditions.	No export without valid V2X authorization; direction transition remains stable and current-limited.	HW/SW	Source-derived	MR-PWR-002

Safety, diagnostics and failure containment

- Bidirectional AFE power flow: No export without valid V2X authorization; direction transition remains stable and current-limited.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-01 - STDES-PFCBIDIR 15 kW three-phase three-level bidirectional AFE (STMicroelectronics): source link
- REF-OBC-02 - TIDA-01606 bidirectional three-phase three-level T-type AFE (Texas Instruments): source link

8. B03 - DC-Link, Precharge, Discharge & HF Decoupling

Intermediate high-voltage bus, film-capacitor energy storage, precharge/discharge and local switching decoupling.

Architecture and boundaries

- DC-link operating window: The internal high-voltage DC link shall be designed around the 700-880 V concept envelope and shall remain within component and control limits across all released modes. Acceptance / boundary: DC-link steady-state/ripple/transient limits close in WCCA and DVP; excursions beyond safe bounds inhibit power.
- Film capacitor life: DC-link film capacitors shall be selected for RMS ripple current, hotspot temperature, voltage transient, lifetime and vibration margin using the charging mission profile. Acceptance / boundary: Lifetime prediction meets program design life with released thermal/ripple profile.
- HF decoupling / busbar ESL: Local HF decoupling and busbar geometry shall limit commutation-loop inductance and device overshoot at released gate-drive settings. Acceptance / boundary: Vds overshoot and ringing remain within device/EMC limits at hot/cold current corners.

Algorithm / dynamic behavior

DC-link supervisor: precharge -> voltage-ratio/dVdt check -> ready -> ripple/OV monitor -> controlled discharge after shutdown.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B03-01	DC-link operating window	The internal high-voltage DC link shall be designed around the 700-880 V concept envelope and shall remain within component and control limits across all released modes.	DC-link steady-state/ripple/transient limits close in WCCA and DVP; excursions beyond safe bounds inhibit power.	HW/SW	Source-derived	MR-EFF-002
OBC30-SYS3-B03-02	Film capacitor life	DC-link film capacitors shall be selected for RMS ripple current, hotspot temperature, voltage transient, lifetime and vibration margin using the charging mission profile.	Lifetime prediction meets program design life with released thermal/ripple profile.	HW	Engineering-derived	MR-LIFE-001
OBC30-SYS3-B03-03	HF decoupling / busbar ESL	Local HF decoupling and busbar geometry shall limit commutation-loop inductance and device overshoot at released gate-drive settings.	Vds overshoot and ringing remain within device/EMC limits at hot/cold current corners.	HW	Engineering-derived	MR-EMC-001
OBC30-SYS3-B03-04	Precharge completion	The controller shall not enable normal AFE/DC-DC switching until the DC-link precharge criteria are met using voltage ratio and plausibility checks.	No full-power enable with incomplete/implausible precharge.	HW/SW	Engineering-derived	MR-PWR-005

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B03-05	Controlled discharge	After shutdown or fault, stored DC-link energy shall be discharged through the released passive/active path and hazardous-voltage state shall be reported until below the released threshold.	Discharge timing and residual voltage meet vehicle/service safety allocation.	HW/SW	Engineering-derived	MR-SRV-001

Safety, diagnostics and failure containment

- DC-link operating window: DC-link steady-state/ripple/transient limits close in WCCA and DVP; excursions beyond safe bounds inhibit power.
- HF decoupling / busbar ESL: Vds overshoot and ringing remain within device/EMC limits at hot/cold current corners.
- Precharge completion: No full-power enable with incomplete/implausible precharge.
- Controlled discharge: Discharge timing and residual voltage meet vehicle/service safety allocation.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-01 - STDES-PFCBIDIR 15 kW three-phase three-level bidirectional AFE (STMicroelectronics): source link
- REF-OBC-03 - STDES-BCBIDIR 11 kW bidirectional battery charger (STMicroelectronics): source link

9. B04 - Isolated Bidirectional DC/DC Modules

Two isolated 15 kW-class bidirectional DAB/CLLC modules connecting the DC link to the 250-920 V battery envelope.

Architecture and boundaries

- Two isolated DC/DC modules: The isolated stage shall provide two 15 kW-class bidirectional conversion channels or equivalent redundancy/modularity. Acceptance / boundary: 30 kW total at released nominal condition; current sharing and degraded operation defined.
- Battery voltage envelope: The DC/DC stage shall support the source concept battery envelope of 250-920 V, including 400 V and 800 V platform variants. Acceptance / boundary: Regulation/protection verified across 250-920 V where the variant is released.
- DAB/CLLC topology control: DAB and CLLC are approved candidate topology classes; final selection shall close efficiency, ZVS range, magnetics, control complexity, isolation capacitance and bidirectional behavior. Acceptance / boundary: Selection rationale and validation evidence linked before design freeze.

Algorithm / dynamic behavior

DC/DC regulator: battery target -> power/current limit -> phase-shift or frequency command -> current-share correction -> ZVS/soft-switch boundary manager.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B04-01	Two isolated DC/DC modules	The isolated stage shall provide two 15 kW-class bidirectional conversion channels or equivalent redundancy/modularity.	30 kW total at released nominal condition; current sharing and degraded operation defined.	HW/SW	Source-derived	MR-PWR-001
OBC30-SYS3-B04-02	Battery voltage envelope	The DC/DC stage shall support the source concept battery envelope of 250-920 V, including 400 V and 800 V platform variants.	Regulation/ protection verified across 250-920 V where the variant is released.	HW/SW	Source-derived	MR-PWR-001
OBC30-SYS3-B04-03	DAB/CLLC topology control	DAB and CLLC are approved candidate topology classes; final selection shall close efficiency, ZVS range, magnetics, control complexity, isolation capacitance and bidirectional behavior.	Selection rationale and validation evidence linked before design freeze.	HW/SW	Source-derived	MR-EFF-001
OBC30-SYS3-B04-04	Isolation barrier	Transformer, PCB and isolation components shall meet reinforced/basic insulation allocations, creepage/clearance and partial-discharge requirements of the released insulation coordination.	No reliance on generic spacing; released WCCA/PD/hipot plan closes for max working/transient voltage.	HW	Engineering-derived	MR-ENV-001
OBC30-SYS3-B04-05	Module current sharing	Parallel DC/DC channels shall share power within a calibrated band and prevent one channel from exceeding thermal/electrical limits during mismatch or derating.	No overload from current-share error; faulted channel is isolated/disabled as allocated.	HW/SW	Engineering-derived	MR-LIFE-001

Safety, diagnostics and failure containment

- Battery voltage envelope: Regulation/protection verified across 250-920 V where the variant is released.
- Isolation barrier: No reliance on generic spacing; released WCCA/PD/hipot plan closes for max working/transient voltage.
- Module current sharing: No overload from current-share error; faulted channel is isolated/disabled as allocated.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-03 - STDES-BCBIDIR 11 kW bidirectional battery charger (STMicroelectronics): source link
- REF-OBC-04 - UM3198 25 kW DAB bidirectional power converter (STMicroelectronics): source link

10. B05 - Gate Drivers & Fast Power-Stage Protection

Isolated SiC gate drive with UVLO, DESAT/OCP, soft-off, Miller control, blanking and hardware fault reporting.

Architecture and boundaries

- Gate-driver isolation and CMTI: SiC gate drivers shall provide the released isolation class and CMTI margin for the measured switching dv/dt environment. Acceptance / boundary: No spurious gate event or fault-reset corruption during worst-case dv/dt / common-mode transient tests.
- Fast DESAT/OCP: Each protected switch/channel shall implement hardware overcurrent/DESAT detection with bounded blanking and soft-off/hard-off behavior. Acceptance / boundary: Fault reaction prevents destructive current/voltage stress within device/module short-circuit capability.
- False-turn-on control: Miller clamp, gate resistance, negative bias if used and layout shall be selected from measured device/module switching evidence rather than a fixed generic recipe. Acceptance / boundary: Gate waveforms remain inside released Vgs/Vds limits with adequate false-turn-on margin.

Algorithm / dynamic behavior

Fast fault path: DESAT/OCP/UVLO -> hardware latch -> gate off / soft-off -> disable permit -> fault status to MCU; recovery only through controlled state.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B05-01	Gate-driver isolation and CMTI	SiC gate drivers shall provide the released isolation class and CMTI margin for the measured switching dv/dt environment.	No spurious gate event or fault-reset corruption during worst-case dv/dt / common-mode transient tests.	HW	Engineering-derived	MR-EMC-001
OBC30-SYS3-B05-02	Fast DESAT/OCP	Each protected switch/channel shall implement hardware overcurrent/DESAT detection with bounded blanking and soft-off/hard-off behavior.	Fault reaction prevents destructive current/voltage stress within device/module short-circuit capability.	HW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B05-03	False-turn-on control	Miller clamp, gate resistance, negative	Gate waveforms remain inside	HW	Engineering-derived	MR-EMC-001

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		bias if used and layout shall be selected from measured device/module switching evidence rather than a fixed generic recipe.	released Vgs/Vds limits with adequate false-turn-on margin.			
OBC30-SYS3-B05-04	Power-up safe state	Gate outputs shall default OFF during reset, brownout, isolated-supply UVLO and MCU unpowered conditions.	No uncontrolled power-device turn-on for any normal power sequencing or defined single fault.	HW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B05-05	Driver fault telemetry	Driver faults shall be latched/reported to the safety/control MCU with channel identity and shall not depend on software to initiate the first hardware shutdown.	Initial trip is hardware autonomous; software captures cause and controls recovery.	HW/SW	Engineering-derived	MR-SRV-001

Safety, diagnostics and failure containment

- Gate-driver isolation and CMTI: No spurious gate event or fault-reset corruption during worst-case dv/dt / common-mode transient tests.
- Fast DESAT/OCP: Fault reaction prevents destructive current/voltage stress within device/module short-circuit capability.
- False-turn-on control: Gate waveforms remain inside released Vgs/Vds limits with adequate false-turn-on margin.
- Power-up safe state: No uncontrolled power-device turn-on for any normal power sequencing or defined single fault.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-05 - Automotive isolated gate-driver portfolio for SiC (Infineon): source link

11. B06 - Voltage, Current & Temperature Sensing

Grid, DC-link, battery and thermal sensing with calibrated accuracy, isolation and plausibility.

Architecture and boundaries

- Grid current sensing: Phase-current sensing shall support PFC control accuracy and protection bandwidth across released current/temperature range. Acceptance / boundary: Error budget supports PF/THD targets; saturation/overrange is detected.

- Grid voltage sensing: AC phase/line voltage sensing shall support PLL, power calculation, grid validity and protection with isolation/spacing appropriate to the topology. Acceptance / boundary: Grid angle and protection thresholds remain within allocated error budget.
- DC-link and battery voltage sensing: DC-link and battery voltage channels shall be independently range-checked and calibrated over temperature; safety thresholds shall not rely solely on one high-precision software path. Acceptance / boundary: Voltage WCCA closes and hazardous OV/UV can force power inhibit.

Algorithm / dynamic behavior

Measurement qualifier: sample -> calibrate -> range/rate/freshness -> redundant plausibility -> publish valid signal or conservative invalid state.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B06-01	Grid current sensing	Phase-current sensing shall support PFC control accuracy and protection bandwidth across released current/temperature range.	Error budget supports PF/THD targets; saturation/overrange is detected.	HW/SW	Engineering-derived	MR-PWR-001
OBC30-SYS3-B06-02	Grid voltage sensing	AC phase/line voltage sensing shall support PLL, power calculation, grid validity and protection with isolation/spacing appropriate to the topology.	Grid angle and protection thresholds remain within allocated error budget.	HW/SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B06-03	DC-link and battery voltage sensing	DC-link and battery voltage channels shall be independently range-checked and calibrated over temperature; safety thresholds shall not rely solely on one high-precision software path.	Voltage WCCA closes and hazardous OV/UV can force power inhibit.	HW/SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B06-04	Battery current / power plausibility	The OBC shall cross-check commanded power against measured battery-side current/voltage and available BMS limits.	Implausible power flow causes controlled reduction or shutdown without optimistic fallback.	HW/SW	Engineering-derived	MR-PWR-004
OBC30-SYS3-B06-05	Temperature sensing	Power semiconductors, magnetics, capacitors and coolant-critical nodes shall have measured or modelled temperatures sufficient for derating and fault protection.	No released 30 kW operating point without thermal observability and margin evidence.	HW/SW	Engineering-derived	MR-ENV-001

Safety, diagnostics and failure containment

- Grid voltage sensing: Grid angle and protection thresholds remain within allocated error budget.

- DC-link and battery voltage sensing: Voltage WCCA closes and hazardous OV/UV can force power inhibit.
- Battery current / power plausibility: Implausible power flow causes controlled reduction or shutdown without optimistic fallback.
- Temperature sensing: No released 30 kW operating point without thermal observability and margin evidence.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-02 - TIDA-01606 bidirectional three-phase three-level T-type AFE (Texas Instruments): source link
- REF-OBC-03 - STDES-BCBIDIR 11 kW bidirectional battery charger (STMicroelectronics): source link

12. B07 - Control MCU, PMIC, Power Tree & Timing Supervision

Automotive real-time controller, PMIC/watchdog, PWM timing, ADC triggering, reset and safe-output supervision.

Architecture and boundaries

- Real-time controller capability: The controller shall provide synchronized high-resolution PWM/ADC resources and computational headroom for two AFE channels, two DC/DC channels and safety/communication tasks. Acceptance / boundary: Worst-case CPU/PWM/ADC timing closes with margin at full feature set.
- PMIC/watchdog supervision: A PMIC or equivalent independent supervisor shall monitor rails/reset/watchdog and enforce safe output defaults. Acceptance / boundary: Missed/early watchdog service, rail faults and reset events inhibit PWM as allocated.
- Timing/WCET: Fast control loops, PWM updates, ADC sampling and safety tasks shall have measured WCET and scheduling margins. Acceptance / boundary: No deadline overrun at worst-case interrupt/DMA/comms load.

Algorithm / dynamic behavior

Startup supervisor: LV power -> PMIC -> reset -> BIST/ECC -> clock/watchdog -> safe PWM initialization -> peripheral self-test -> enable charging state machine.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B07-01	Real-time controller capability	The controller shall provide synchronized high-resolution PWM/ADC resources and computational headroom for two AFE channels, two DC/DC channels and safety/communication tasks.	Worst-case CPU/PWM/ADC timing closes with margin at full feature set.	HW/SW	Engineering-derived	MR-PWR-001
OBC30-SYS3-	PMIC/watchdog	A PMIC or	Missed/early	HW/SW	Engineering-	MR-PWR-005

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
B07-02	supervision	equivalent independent supervisor shall monitor rails/reset/watchdog and enforce safe output defaults.	watchdog service, rail faults and reset events inhibit PWM as allocated.		derived	
OBC30-SYS3-B07-03	Timing/WCET	Fast control loops, PWM updates, ADC sampling and safety tasks shall have measured WCET and scheduling margins.	No deadline overrun at worst-case interrupt/DMA/comms load.	SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B07-04	Memory/clock safety	ECC/parity, clock monitoring and memory protection shall be enabled where supported and mapped to a defined reaction.	Injected clock/memory faults are detected and lead to safe/degraded state within allocated FTTI.	HW/SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B07-05	Startup self-test	Startup shall execute controller/PMIC/peripheral/gate-drive self-tests before enabling high-voltage switching.	No power conversion before critical self-test completion and valid authorizations.	SW	Engineering-derived	MR-PWR-005

Safety, diagnostics and failure containment

- Real-time controller capability: Worst-case CPU/PWM/ADC timing closes with margin at full feature set.
- PMIC/watchdog supervision: Missed/early watchdog service, rail faults and reset events inhibit PWM as allocated.
- Timing/WCET: No deadline overrun at worst-case interrupt/DMA/comms load.
- Memory/clock safety: Injected clock/memory faults are detected and lead to safe/degraded state within allocated FTTI.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-06 - ST 7 kW OBC reference design with automotive MCU (STMicroelectronics): source link

13. B08 - Grid Synchronization & PFC Control

Grid angle/frequency tracking, dq-current regulation, PF/THD control and bidirectional current reference generation.

Architecture and boundaries

- Grid synchronization: The PFC controller shall track valid grid phase/frequency with a bounded PLL/estimator and shall declare invalid state for severe distortion/out-of-range conditions. Acceptance / boundary: Stable angle tracking through released grid distortion/sag envelope; invalid grid blocks export/charge as allocated.

- dq current control: The AFE shall regulate dq currents with decoupling/feed-forward and saturation handling adequate to meet PF/THD and dynamic power commands. Acceptance / boundary: No integrator windup or control instability at voltage/current limits.
- Reactive current policy: Iq/reactive-power command shall default to the released charging policy and shall not be independently commanded outside a separately approved V2G/grid-service function. Acceptance / boundary: No unauthorized reactive power export.

Algorithm / dynamic behavior

PLL/VOC loop: validate grid -> estimate angle -> generate Id/Iq reference -> current PI/PR control -> feed-forward/decoupling -> PWM duty update.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B08-01	Grid synchronization	The PFC controller shall track valid grid phase/frequency with a bounded PLL/estimator and shall declare invalid state for severe distortion/out-of-range conditions.	Stable angle tracking through released grid distortion/sag envelope; invalid grid blocks export/charge as allocated.	SW	Engineering-derived	MR-PWR-001
OBC30-SYS3-B08-02	dq current control	The AFE shall regulate dq currents with decoupling/feed-forward and saturation handling adequate to meet PF/THD and dynamic power commands.	No integrator windup or control instability at voltage/current limits.	SW	Engineering-derived	MR-PWR-001
OBC30-SYS3-B08-03	Reactive current policy	Iq/reactive-power command shall default to the released charging policy and shall not be independently commanded outside a separately approved V2G/grid-service function.	No unauthorized reactive power export.	SW	Engineering-derived	MR-PWR-002
OBC30-SYS3-B08-04	Direction transition	Grid-to-battery and battery-to-grid transition shall ramp current/power through zero with controlled sign change.	No uncontrolled DC-link transient or current step beyond released limit.	SW	Engineering-derived	MR-PWR-002
OBC30-SYS3-B08-05	Control mode freeze	Every released PWM/modulation mode shall have its own efficiency/EMC/control-stability evidence; unvalidated lab modes shall be blocked in production calibration.	Production calibration cannot enable unreleased switching mode.	SW	Engineering-derived	MR-EMC-001

Safety, diagnostics and failure containment

- Grid synchronization: Stable angle tracking through released grid distortion/sag envelope; invalid grid blocks export/charge as allocated.
- Reactive current policy: No unauthorized reactive power export.
- Direction transition: No uncontrolled DC-link transient or current step beyond released limit.
- Control mode freeze: Production calibration cannot enable unreleased switching mode.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-01 - STDES-PFCBIDIR 15 kW three-phase three-level bidirectional AFE (STMicroelectronics): source link
- REF-OBC-02 - TIDA-01606 bidirectional three-phase three-level T-type AFE (Texas Instruments): source link

14. B09 - DC/DC Modulation, Power Flow & Current Sharing Control

DAB/CLLC modulation, bus/battery regulation, module balancing and direction changes.

Architecture and boundaries

- Power command conversion: The DC/DC controller shall convert requested battery/grid power to bounded module current/power references using measured voltages and direction. Acceptance / boundary: References respect BMS/EVSE/thermal/hardware limits.
- DAB/CLLC modulation: The selected modulation algorithm shall keep commanded duty/phase/frequency inside transformer, switch, resonant and ZVS limits. Acceptance / boundary: No operation in a released point that violates device/magnetics/current limits.
- Current-share loop: Module power/current share shall use a slower trim loop that does not destabilize the primary voltage/current control loops. Acceptance / boundary: Channel mismatch remains within calibrated band without low-frequency hunting.

Algorithm / dynamic behavior

Power-flow manager: requested power -> module availability -> split command -> modulation command -> current share trim -> saturation/soft-switch protection.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B09-01	Power command conversion	The DC/DC controller shall convert requested battery/grid power to bounded module current/power references using measured voltages and direction.	References respect BMS/EVSE/thermal/hardware limits.	SW	Engineering-derived	MR-PWR-004

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B09-02	DAB/CLLC modulation	The selected modulation algorithm shall keep commanded duty/phase/frequency inside transformer, switch, resonant and ZVS limits.	No operation in a released point that violates device/magnetics/current limits.	SW	Engineering-derived	MR-EFF-001
OBC30-SYS3-B09-03	Current-share loop	Module power/current share shall use a slower trim loop that does not destabilize the primary voltage/current control loops.	Channel mismatch remains within calibrated band without low-frequency hunting.	SW	Engineering-derived	MR-EFF-002
OBC30-SYS3-B09-04	Bumpless mode change	Changes among startup, charge, V2X, derate and shutdown modulation regimes shall be bumpless or explicitly ramped.	No step causing overcurrent, transformer saturation or DC-link excursion.	SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B09-05	Soft-switching efficiency manager	Where ZVS/ZCS boundaries are actively managed, optimization shall never raise power above deterministic thermal/electrical/safety limits.	Efficiency optimization is subordinate to hard limits.	SW	Engineering-derived	MR-EFF-001

Safety, diagnostics and failure containment

- Power command conversion: References respect BMS/EVSE/thermal/hardware limits.
- DAB/CLLC modulation: No operation in a released point that violates device/magnetics/current limits.
- Bumpless mode change: No step causing overcurrent, transformer saturation or DC-link excursion.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-03 - STDES-BCBIDIR 11 kW bidirectional battery charger (STMicroelectronics): source link
- REF-OBC-04 - UM3198 25 kW DAB bidirectional power converter (STMicroelectronics): source link

15. B10 - Charging State Machine & Power Arbitration

End-to-end operating states, BMS/EVSE authorization, power limit arbitration, start/stop and fault recovery.

Architecture and boundaries

- Operating state machine: The OBC shall implement controlled OFF, WAKE, SELF-TEST, GRID-VALID, PRECHARGE, READY, CHARGE/V2X, DERATE, SHUTDOWN and FAULT states. Acceptance / boundary: Every transition has defined prerequisites, outputs, timeout and recovery behavior.
- Power authorization: Charging/export shall require valid vehicle/BMS/EVSE authorization and valid electrical/thermal/safety prerequisites. Acceptance / boundary: Loss of any required authorization causes bounded reduction to zero or immediate inhibit per severity.
- Most restrictive limit: The power arbitration manager shall apply the minimum valid power/current limit from EVSE, BMS, thermal, hardware and regional constraints. Acceptance / boundary: No stale/invalid source can increase allowed power.

Algorithm / dynamic behavior

OFF -> WAKE -> SELF-TEST -> GRID VALID -> PRECHARGE -> READY -> CHARGE/V2X -> DERATE -> SHUTDOWN/FAULT; most restrictive valid power limit always wins.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B10-01	Operating state machine	The OBC shall implement controlled OFF, WAKE, SELF-TEST, GRID-VALID, PRECHARGE, READY, CHARGE/V2X, DERATE, SHUTDOWN and FAULT states.	Every transition has defined prerequisites, outputs, timeout and recovery behavior.	SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B10-02	Power authorization	Charging/export shall require valid vehicle/BMS/EVSE authorization and valid electrical/thermal/safety prerequisites.	Loss of any required authorization causes bounded reduction to zero or immediate inhibit per severity.	SW	Source-derived	MR-PWR-004
OBC30-SYS3-B10-03	Most restrictive limit	The power arbitration manager shall apply the minimum valid power/current limit from EVSE, BMS, thermal, hardware and regional constraints.	No stale/invalid source can increase allowed power.	SW	Source-derived	MR-PWR-004
OBC30-SYS3-B10-04	Authorization removal	On charge/export authorization removal, commanded power shall transition to zero with no uncontrolled reverse energy.	Power crosses to zero within released ramp/emergency timing; contactor/gate sequence remains safe.	HW/SW	Source-derived	MR-PWR-005
OBC30-SYS3-B10-05	Fault retry policy	Automatic retries after grid, thermal, isolation or power-stage faults shall be bounded by fault class, cooldown/energy limits and retry counter.	No rapid repetitive stressing of contactors, precharge paths, magnetics or power semiconductors.	SW	Engineering-derived	MR-LIFE-001

Safety, diagnostics and failure containment

- Operating state machine: Every transition has defined prerequisites, outputs, timeout and recovery behavior.
- Power authorization: Loss of any required authorization causes bounded reduction to zero or immediate inhibit per severity.
- Most restrictive limit: No stale/invalid source can increase allowed power.
- Authorization removal: Power crosses to zero within released ramp/emergency timing; contactor/gate sequence remains safe.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-07 - ISO 15118 road vehicles - vehicle-to-grid communication interface (ISO): source link
- REF-OBC-08 - CharIN charging ecosystem and ISO 15118 resources (CharIN): source link

16. B11 - Thermal Management & Derating

Vehicle liquid-loop interface, hotspot monitoring, temperature estimation and monotonic derating.

Architecture and boundaries

- Liquid cooling interface: The OBC shall use the vehicle liquid loop and shall monitor coolant availability/state needed to sustain the released power level. Acceptance / boundary: 30 kW operation only when coolant conditions are within released envelope.
- Monotonic thermal derating: Charge/export limits shall decrease monotonically as semiconductor, magnetics, capacitor or coolant limits are approached. Acceptance / boundary: Hotter or invalid data never increases power capability.
- Recovery hysteresis: Thermal power recovery shall use calibrated hysteresis and a slower upward rate to avoid oscillation with EVSE/BMS/vehicle power loops. Acceptance / boundary: No repeated power hunting around a thermal threshold.

Algorithm / dynamic behavior

Thermal manager: sensor validity -> hotspot/max -> coolant status -> loss/temperature model -> derate limit -> hysteretic slow recovery -> shutdown if hard limit.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B11-01	Liquid cooling interface	The OBC shall use the vehicle liquid loop and shall monitor coolant availability/state needed to sustain the released power level.	30 kW operation only when coolant conditions are within released envelope.	HW/SW	Source-derived	MR-ENV-001
OBC30-SYS3-	Monotonic thermal	Charge/export	Hotter or invalid	SW	Engineering-	MR-ENV-001

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
B11-02	derating	limits shall decrease monotonically as semiconductor, magnetics, capacitor or coolant limits are approached.	data never increases power capability.		derived	
OBC30-SYS3-B11-03	Recovery hysteresis	Thermal power recovery shall use calibrated hysteresis and a slower upward rate to avoid oscillation with EVSE/BMS/vehicle power loops.	No repeated power hunting around a thermal threshold.	SW	Engineering-derived	MR-ENV-001
OBC30-SYS3-B11-04	Coolant-loss reaction	Loss of required coolant flow/temperature plausibility shall trigger a bounded derate or shutdown before component thermal limits are exceeded.	Thermal model/bench test demonstrates margin at worst-case stored heat.	HW/SW	Engineering-derived	MR-ENV-001
OBC30-SYS3-B11-05	Thermal model correlation	Any virtual junction/hotspot estimator used for limiting shall be correlated to instrumented hardware across representative loss distributions.	Model error remains within released calibration band; model cannot raise hard safety limits.	SW	Engineering-derived	MR-LIFE-001

Safety, diagnostics and failure containment

- Liquid cooling interface: 30 kW operation only when coolant conditions are within released envelope.
- Monotonic thermal derating: Hotter or invalid data never increases power capability.
- Coolant-loss reaction: Thermal model/bench test demonstrates margin at worst-case stored heat.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-03 - STDES-BCBIDIR 11 kW bidirectional battery charger (STMicroelectronics): source link

17. B12 - Functional Safety, HVIL, IMD & Independent Inhibit

Independent safety prerequisites and hardware/software reaction paths for unsafe HV or power-conversion conditions.

Architecture and boundaries

- Safe state definition: The OBC safety concept shall define hardware/software safe states for AC-side faults, DC-side faults, isolation faults and controller faults. Acceptance / boundary: Safe-state outputs and energy-discharge behavior trace to HARA/FSC/TSC.
- HVIL/vehicle inhibit: Vehicle HV inhibit/crash/service inputs shall remove power-stage permission through an independent or sufficiently diverse path where allocated. Acceptance / boundary: Fault injection demonstrates gate inhibit independent of normal application command.
- IMD/isolation fault: Invalid isolation-monitor status or diagnosed insulation failure shall prevent or terminate charging/export according to vehicle safety allocation. Acceptance / boundary: No continued high-power operation with invalid mandatory isolation status.

Algorithm / dynamic behavior

Safety arbiter: watchdog + HVIL + IMD + driver faults + sensor plausibility + authorization -> permit/inhibit power stage -> safe-state selection.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B12-01	Safe state definition	The OBC safety concept shall define hardware/software safe states for AC-side faults, DC-side faults, isolation faults and controller faults.	Safe-state outputs and energy-discharge behavior trace to HARA/FSC/TSC.	HW/SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B12-02	HVIL/vehicle inhibit	Vehicle HV inhibit/crash/service inputs shall remove power-stage permission through an independent or sufficiently diverse path where allocated.	Fault injection demonstrates gate inhibit independent of normal application command.	HW/SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B12-03	IMD/isolation fault	Invalid isolation-monitor status or diagnosed insulation failure shall prevent or terminate charging/export according to vehicle safety allocation.	No continued high-power operation with invalid mandatory isolation status.	HW/SW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B12-04	Independent severe-fault path	Severe overcurrent/overvoltage power-stage faults shall have a reaction path that does not rely solely on the main application task or network communication.	Hardware fault path acts inside device/system FTTI.	HW	Engineering-derived	MR-PWR-005
OBC30-SYS3-B12-05	Safety diagnostics	Safety mechanisms shall be exercised at startup and/or periodically at intervals justified by latent-fault analysis.	Diagnostic coverage/interval linked to FMEDA/HARA allocation.	HW/SW	Engineering-derived	MR-PWR-005

Safety, diagnostics and failure containment

- Safe state definition: Safe-state outputs and energy-discharge behavior trace to HARA/FSC/TSC.
- HVIL/vehicle inhibit: Fault injection demonstrates gate inhibit independent of normal application command.
- IMD/isolation fault: No continued high-power operation with invalid mandatory isolation status.
- Independent severe-fault path: Hardware fault path acts inside device/system FTTI.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-GEN-01 - ISO 26262 road vehicles - functional safety (ISO): source link

18. B13 - Charging Communications: CAN FD, EVCC/PLC & ISO 15118

Vehicle/BMS communication plus EVSE high-level communication and bidirectional authorization.

Architecture and boundaries

- CAN FD vehicle interface: The OBC shall exchange state, limits, faults and commands with BMS/VCU using the released CAN FD/E2E interface. Acceptance / boundary: Critical message freshness, CRC/counter and timeout behavior verified.
- EVCC/PLC interface: High-level charging communication shall interface to the released EVCC/PLC architecture and shall not permit power before communication prerequisites are satisfied. Acceptance / boundary: Charging/export enable follows released charging protocol state.
- ISO 15118 / V2X authorization: Bidirectional power export shall be enabled only for released vehicle/market configurations and after successful required protocol/security authorization. Acceptance / boundary: No V2X power on unsupported market/configuration or failed authorization.

Algorithm / dynamic behavior

Comms manager: receive -> E2E/freshness/authentication -> range/plausibility -> application handoff; stale/invalid critical messages force conservative power or stop.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B13-01	CAN FD vehicle interface	The OBC shall exchange state, limits, faults and commands with BMS/VCU using the released CAN FD/E2E interface.	Critical message freshness, CRC/counter and timeout behavior verified.	HW/SW	Engineering-derived	MR-PWR-004
OBC30-SYS3-B13-02	EVCC/PLC interface	High-level charging communication shall interface to the released EVCC/PLC	Charging/export enable follows released charging protocol state.	HW/SW	Engineering-derived	MR-PWR-002

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		architecture and shall not permit power before communication prerequisites are satisfied.				
OBC30-SYS3-B13-03	ISO 15118 / V2X authorization	Bidirectional power export shall be enabled only for released vehicle/market configurations and after successful required protocol/security authorization.	No V2X power on unsupported market/configuration or failed authorization.	SW	Engineering-derived	MR-PWR-002
OBC30-SYS3-B13-04	Message timeout	Stale or invalid critical BMS/EVSE limits shall force conservative power limitation or shutdown, never a higher fallback.	Timeout reactions verified by HIL/network fault injection.	SW	Engineering-derived	MR-PWR-004
OBC30-SYS3-B13-05	Comms fault containment	Diagnostic/service traffic shall not starve safety/control communication or violate real-time task deadlines.	Bus-load and DoS-style stress preserves safety/control timing.	HW/SW	Engineering-derived	MR-CYB-001

Safety, diagnostics and failure containment

- CAN FD vehicle interface: Critical message freshness, CRC/counter and timeout behavior verified.
- EVCC/PLC interface: Charging/export enable follows released charging protocol state.
- ISO 15118 / V2X authorization: No V2X power on unsupported market/configuration or failed authorization.
- Message timeout: Timeout reactions verified by HIL/network fault injection.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-07 - ISO 15118 road vehicles - vehicle-to-grid communication interface (ISO): source link
- REF-OBC-08 - CharIN charging ecosystem and ISO 15118 resources (CharIN): source link

19. B14 - Diagnostics, UDS, NVM & Service Logging

DTCs, freeze frames, operating counters, service routines and persistent fault evidence.

Architecture and boundaries

- DTC/freeze frame: The OBC shall store diagnosable fault codes with timestamped freeze-frame context for power-stage, thermal, communication, isolation and safety faults. Acceptance / boundary: Required service data survives power cycle within NVM endurance plan.
- Operating counters: Energy throughput, charge/export hours, thermal events and major fault counters shall be retained for service/warranty analysis. Acceptance / boundary: Counter integrity/endurance defined; rollover behavior controlled.
- UDS service routines: Service/calibration routines shall require a controlled state and default to zero power / disabled switching when invasive tests are active. Acceptance / boundary: No unintended power generation during diagnostic session.

Algorithm / dynamic behavior

Diagnostic manager: debounce/qualify -> DTC -> freeze-frame/counter -> persistent record -> service policy -> healing/aging; safety path never depends on logging.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B14-01	DTC/freeze frame	The OBC shall store diagnosable fault codes with timestamped freeze-frame context for power-stage, thermal, communication, isolation and safety faults.	Required service data survives power cycle within NVM endurance plan.	SW	Source-derived	MR-SRV-001
OBC30-SYS3-B14-02	Operating counters	Energy throughput, charge/export hours, thermal events and major fault counters shall be retained for service/warranty analysis.	Counter integrity/endurance defined; rollover behavior controlled.	SW	Source-derived	MR-SRV-001
OBC30-SYS3-B14-03	UDS service routines	Service/calibration routines shall require a controlled state and default to zero power / disabled switching when invasive tests are active.	No unintended power generation during diagnostic session.	SW	Engineering-derived	MR-SRV-001
OBC30-SYS3-B14-04	Flight recorder	High-rate rolling event capture shall record selected grid/DC/battery/control variables around critical trips where memory budget permits.	Pre/post trigger evidence available for qualified fault classes.	SW	Engineering-derived	MR-SRV-001
OBC30-SYS3-B14-05	Logging independence	Failure or saturation of logging/NVM shall not block an immediate safety reaction.	Safety shutdown remains functional under NVM/logging fault injection.	SW	Engineering-derived	MR-SRV-001

Safety, diagnostics and failure containment

- UDS service routines: No unintended power generation during diagnostic session.
- Logging independence: Safety shutdown remains functional under NVM/logging fault injection.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-GEN-03 - ISO 14229 unified diagnostic services (ISO): source link

20. B15 - Cybersecurity, Secure Boot, Update & Calibration

Root of trust, authenticated update, rollback protection, calibration integrity and variant compatibility.

Architecture and boundaries

- Secure boot: Production software shall be authenticated before execution and shall keep power-stage enable inhibited on authentication failure. Acceptance / boundary: Unsigned/modified production image blocked.
- Authenticated update / rollback: Software update shall verify authenticity/integrity/version and recover safely from interrupted programming without unauthorized rollback. Acceptance / boundary: Interrupted/invalid update does not enable power conversion.
- Calibration authenticity and hard bounds: Safety/power calibration shall be signed/authenticated and additionally constrained by immutable hard ranges. Acceptance / boundary: Valid signature alone cannot authorize out-of-bound values.

Algorithm / dynamic behavior

Secure start/update: authenticate image -> version/rollback -> calibration signature/range/variant -> application handover; failure keeps power stage inhibited.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B15-01	Secure boot	Production software shall be authenticated before execution and shall keep power-stage enable inhibited on authentication failure.	Unsigned/modified production image blocked.	SW	Source-derived	MR-CYB-001
OBC30-SYS3-B15-02	Authenticated update / rollback	Software update shall verify authenticity/integrity/version and recover safely from interrupted programming without unauthorized rollback.	Interrupted/invalid update does not enable power conversion.	SW	Engineering-derived	MR-CYB-001
OBC30-SYS3-B15-03	Calibration authenticity and hard bounds	Safety/power calibration shall be signed/authenticated and additionally constrained by	Valid signature alone cannot authorize out-of-bound values.	SW	Engineering-derived	MR-CYB-001

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		immutable hard ranges.				
OBC30-SYS3-B15-04	Variant coding	Calibration/ application variant ID shall match physical OBC voltage class, topology/module configuration and vehicle program before power-stage enable.	Wrong-variant signed package is rejected.	HW/SW	Engineering-derived	MR-CYB-001
OBC30-SYS3-B15-05	Key/certificate protection	Security keys and charging certificates shall use approved HSM/secure storage and access-control lifecycle.	Unauthorized read/write attempts blocked; manufacturing provisioning traceable.	HW/SW	Engineering-derived	MR-CYB-001

Safety, diagnostics and failure containment

- Secure boot: Unsigned/modified production image blocked.
- Authenticated update / rollback: Interrupted/invalid update does not enable power conversion.
- Calibration authenticity and hard bounds: Valid signature alone cannot authorize out-of-bound values.
- Variant coding: Wrong-variant signed package is rejected.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-GEN-02 - ISO/SAE 21434 road vehicles - cybersecurity engineering (ISO): source link

21. B16 - Mechanical, EMC, Isolation, Grounding & NVH

Enclosure, shielding, grounding, insulation coordination, coolant sealing, vibration and acoustic behavior.

Architecture and boundaries

- Insulation coordination: HV-LV/chassis insulation, creepage and clearance shall be derived from maximum working/transient voltage, pollution/material class and OEM/regulatory requirements. Acceptance / boundary: Released WCCA/PD/hipot plan; no generic spacing assumption.
- Grounding/shielding: Chassis, LV ground, shields, Y-capacitors and common-mode paths shall be intentionally controlled to balance EMC and leakage-current constraints. Acceptance / boundary: EMC/leakage/isolation tests pass in every released switching mode.
- Environmental robustness: Enclosure, PCB, connectors and cooling interfaces shall survive released temperature, vibration, humidity, salt, dust, water and altitude profiles. Acceptance / boundary: No unsafe degradation after environmental DVP.

Algorithm / dynamic behavior

Not a real-time algorithm; design governance links mechanical/EMC/insulation special characteristics to release tests and change control.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B16-01	Insulation coordination	HV-LV/chassis insulation, creepage and clearance shall be derived from maximum working/transient voltage, pollution/material class and OEM/regulatory requirements.	Released WCCA/PD/hipot plan; no generic spacing assumption.	HW	Engineering-derived	MR-ENV-001
OBC30-SYS3-B16-02	Grounding/ shielding	Chassis, LV ground, shields, Y-capacitors and common-mode paths shall be intentionally controlled to balance EMC and leakage-current constraints.	EMC/leakage/ isolation tests pass in every released switching mode.	HW	Engineering-derived	MR-EMC-001
OBC30-SYS3-B16-03	Environmental robustness	Enclosure, PCB, connectors and cooling interfaces shall survive released temperature, vibration, humidity, salt, dust, water and altitude profiles.	No unsafe degradation after environmental DVP.	HW	Source-derived	MR-ENV-001
OBC30-SYS3-B16-04	Mass and volume targets	Mechanical architecture shall track <=15 kg and <=12 L program targets while separately reporting active power-stage volume.	Mass/volume measured at design gates; deviations change-controlled.	HW	Source-derived	MR-MASS-001;MR-VOL-001
OBC30-SYS3-B16-05	NVH	Magnetics, switching-frequency strategy and mechanical mounting shall meet the stationary charging audible tonal-noise target.	Semi-anechoic test passes vehicle NVH limit across released load/mode range.	HW/SW	Source-derived	MR-NVH-001

Safety, diagnostics and failure containment

- Insulation coordination: Released WCCA/PD/hipot plan; no generic spacing assumption.
- Grounding/shielding: EMC/leakage/isolation tests pass in every released switching mode.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner

- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-05 - Automotive isolated gate-driver portfolio for SiC (Infineon): source link

22. B17 - Reliability, Design Life, Derating & 150% BOM

Mission-profile lifetime, component stress derating, supply continuity and alternate technology evidence.

Architecture and boundaries

- Design life: Power semiconductors, capacitors, magnetics, relays/connectors, isolation components and cooling structure shall be assessed against the 15-year/300,000 km target mission profile. Acceptance / boundary: Lifetime/aging margin documented for released electrical/thermal mission.
- Component derating: Voltage/current/temperature/power stress shall use explicit BOL/EOL derating rules and tolerance/aging analysis. Acceptance / boundary: No critical component released at unreviewed absolute-maximum dependence.
- 150% BOM: Strategic components shall have approved alternate technology or sourcing evidence where program policy requires it. Acceptance / boundary: Alternate is not assumed drop-in; interface/safety/EMC/thermal/software impacts documented.

Algorithm / dynamic behavior

Reliability governance: mission profile -> stress/loss model -> lifetime consumption -> margin check -> alternate part impact -> DVP/DRBFM update.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B17-01	Design life	Power semiconductors, capacitors, magnetics, relays/connectors, isolation components and cooling structure shall be assessed against the 15-year/300,000 km target mission profile.	Lifetime/aging margin documented for released electrical/thermal mission.	HW	Source-derived	MR-LIFE-001
OBC30-SYS3-B17-02	Component derating	Voltage/current/temperature/power stress shall use explicit BOL/EOL derating rules and tolerance/aging analysis.	No critical component released at unreviewed absolute-maximum dependence.	HW	Engineering-derived	MR-LIFE-001
OBC30-SYS3-B17-03	150% BOM	Strategic components shall have approved alternate technology or sourcing evidence where program policy requires it.	Alternate is not assumed drop-in; interface/safety/EMC/thermal/software impacts documented.	HW/SW	Engineering-derived	MR-COST-001
OBC30-SYS3-	Supplier change	Silicon/magnetics/	PCN/errata linked	HW/SW	Engineering-	MR-COST-001

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
B17-04	control	capacitor/driver revisions and process changes affecting special characteristics shall trigger impact assessment and regression scope.	to DFMEA/DRBFM/ DVP before release.		derived	
OBC30-SYS3-B17-05	Cost governance	Architecture and BOM shall remain consistent with the program-approved piece-cost target without bypassing safety/reliability/validation requirements.	Cost changes are documented trade-offs, not silent requirement deletion.	SYS	Source-derived	MR-COST-001

Safety, diagnostics and failure containment

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-OBC-09 - Wolfspeed 25 kW high-efficiency bidirectional T-type converter (Wolfspeed): source link

23. B18 - Manufacturing, EOL Calibration & Traceability

Automated EOL electrical/insulation/comms/leak tests, calibration, serialized genealogy and release evidence.

Architecture and boundaries

- Automated EOL tests: Production shall support automated electrical, insulation, communication and leak tests using released fixtures and limits. Acceptance / boundary: Every serial has pass/fail record for mandatory EOL tests.
- Calibration trace: Current/voltage/temperature calibration shall record fixture ID, master reference identity, correlation/drift status, SW/cal version and results. Acceptance / boundary: Out-of-correlation fixture blocks product release.
- Serialized genealogy: Unit serial shall link to BOM/lot, power-module identity, PCB revision, software, calibration, security provisioning and EOL results. Acceptance / boundary: Genealogy record complete before shipment release.

Algorithm / dynamic behavior

EOL flow: identity -> flash/security -> calibration -> electrical/insulation/comms/leak tests -> record fixture/reference -> pass/fail lock -> genealogy upload.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
OBC30-SYS3-B18-01	Automated EOL tests	Production shall support automated electrical,	Every serial has pass/fail record for mandatory EOL	HW/SW	Source-derived	MR-MFG-001

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		insulation, communication and leak tests using released fixtures and limits.	tests.			
OBC30-SYS3-B18-02	Calibration trace	Current/voltage/ temperature calibration shall record fixture ID, master reference identity, correlation/drift status, SW/cal version and results.	Out-of-correlation fixture blocks product release.	HW/SW	Engineering-derived	MR-MFG-001
OBC30-SYS3-B18-03	Serialized genealogy	Unit serial shall link to BOM/lot, power-module identity, PCB revision, software, calibration, security provisioning and EOL results.	Genealogy record complete before shipment release.	SYS	Engineering-derived	MR-MFG-001
OBC30-SYS3-B18-04	EOL safe configuration	EOL/service modes shall default to power stage inhibited unless a controlled test explicitly needs energized operation.	No uncontrolled HV switching from production test commands.	HW/SW	Engineering-derived	MR-MFG-001
OBC30-SYS3-B18-05	Release evidence completeness	DV/PV/SOP release shall require traceability to approved procedures, configuration, raw result, pass/fail and deviation closure.	A populated planned-test row is not counted as executed evidence.	SYS	Engineering-derived	MR-MFG-001

Safety, diagnostics and failure containment

- EOL safe configuration: No uncontrolled HV switching from production test commands.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-GEN-04 - IATF 16949 automotive quality management (IATF): source link

24. Verification, DVP and Release Evidence Strategy

- MIL/SIL for control algorithms and limit managers before target integration.
- Target/HIL fault injection for sensor/network/CPU/safety-path timing.
- Bench double-pulse / power-stage / insulation / thermal tests as applicable.
- Power-analyzer/dyno/grid-simulator testing across released operating envelope.

- EMC, environmental, vibration, thermal shock, ingress and durability per vehicle DVP.
- EOL fixture correlation and serialized production evidence before SOP release.

25. Reference Document Register

External references provide design comparators, datasheets/application notes and standards context. They do not supersede the controlled product baseline. Freeze the exact revision/PDF in the project repository before release.

Ref ID	Block	Category	Document / Title	Source	Link	Use
REF-GEN-01	General	Standard	ISO 26262 road vehicles - functional safety	ISO	https://www.iso.org/standard/68383.html	Functional safety lifecycle and safety-mechanism allocation.
REF-GEN-02	General	Standard	ISO/SAE 21434 road vehicles - cybersecurity engineering	ISO	https://www.iso.org/standard/70918.html	Cybersecurity lifecycle and TARA reference.
REF-GEN-03	General	Standard	ISO 14229 unified diagnostic services	ISO	https://www.iso.org/standard/72439.html	UDS service and diagnostic behavior.
REF-GEN-04	General	Other	IATF 16949 automotive quality management	IATF	https://www.iatfglobaloversight.org/iatf-16949/	Manufacturing and production quality context.
REF-OBC-01	B01/B02/B08	Reference Design	STDES-PFCBIDIR 15 kW three-phase three-level bidirectional AFE	STMicroelectronics	https://www.st.com/en/evaluation-tools/stdes-pfcbidir.html	AFE topology, digital control and 800 V / 400 V reference comparator.
REF-OBC-02	B01/B02/B06/B08	Reference Design	TIDA-01606 bidirectional three-phase three-level T-type AFE	Texas Instruments	https://www.ti.com/tool/TIDA-01606	DQ control, T-type AFE, LCL/filter and efficiency comparator.
REF-OBC-03	B03/B04/B11	Reference Design	STDES-BCBIDIR 11 kW bidirectional battery charger	STMicroelectronics	https://www.st.com/en/evaluation-tools/stdes-bcbidir.html	Integrated PFC + isolated DAB/CLLC charger comparator.
REF-OBC-04	B04/B09	User Manual	UM3198 25 kW DAB bidirectional power converter	STMicroelectronics	https://www.st.com/resource/en/user_manual/um3198-25-kw-dual-active-bridge-bidirectional-power-converter-for-ev-charging-and-battery-energy-storage-systems-stmicroelectronics.pdf	High-power DAB control, magnetics and validation reference.
REF-OBC-05	B05/B16	Product / Application	Automotive isolated gate-driver portfolio for SiC	Infineon	https://www.infineon.com/products/power/gate-driver-ics/automotive	Gate isolation, CMTI and traction/OBC device reference.
REF-OBC-06	B07	Application Note	ST 7 kW OBC reference design with automotive MCU	STMicroelectronics	https://www.st.com/resource/en/application_note/an5949-sr5e1-line--7-kw-evhev-onboardcharger-reference-design-stmicroelectronics.pdf	Controller/peripheral/reference software comparator.
REF-OBC-07	B10/B13	Standard	ISO 15118 road vehicles - vehicle-	ISO	https://www.iso.org/	Charging high-level

Ref ID	Block	Category	Document / Title	Source	Link	Use
			to-grid communication interface		standard/ 77845.html	communication reference.
REF-OBC-08	B10/B13	Industry Implementation	CharIN charging ecosystem and ISO 15118 resources	CharIN	https:// www.charin.global /technology/ iso15118/	Implementation/ ecosystem reference; project-specific compliance remains separate.
REF-OBC-09	B17	Reference Design	Wolfspeed 25 kW high-efficiency bidirectional T-type converter	Wolfspeed	https:// www.wolfspeed.com/products/power/ reference-designs/ crd-25bda6512n-k/	SiC high-power bidirectional converter design comparator.

Clickable reference links

- REF-GEN-01 - ISO 26262 road vehicles - functional safety: https://www.iso.org/standard/68383.html
- REF-GEN-02 - ISO/SAE 21434 road vehicles - cybersecurity engineering: https://www.iso.org/standard/70918.html
- REF-GEN-03 - ISO 14229 unified diagnostic services: https://www.iso.org/standard/72439.html
- REF-GEN-04 - IATF 16949 automotive quality management: https://www.iatfglobaloversight.org/iatf-16949/
- REF-OBC-01 - STDES-PFCBIDIR 15 kW three-phase three-level bidirectional AFE: https://www.st.com/en/evaluation-tools/stdes-pfcbidir.html
- REF-OBC-02 - TIDA-01606 bidirectional three-phase three-level T-type AFE: https://www.ti.com/tool/TIDA-01606
- REF-OBC-03 - STDES-BCBIDIR 11 kW bidirectional battery charger: https://www.st.com/en/evaluation-tools/stdes-bcbidir.html
- REF-OBC-04 - UM3198 25 kW DAB bidirectional power converter: https://www.st.com/resource/en/user_manual/um3198-25-kw-dual-active-bridge-bidirectional-power-converter-for-ev-charging-and-battery-energy-storage-systems-stmicroelectronics.pdf
- REF-OBC-05 - Automotive isolated gate-driver portfolio for SiC: https://www.infineon.com/products/power/gate-driver-ics/automotive
- REF-OBC-06 - ST 7 kW OBC reference design with automotive MCU: https://www.st.com/resource/en/application_note/an5949-sr5e1-line--7-kw-evhev-onboardcharger-reference-design-stmicroelectronics.pdf
- REF-OBC-07 - ISO 15118 road vehicles - vehicle-to-grid communication interface: https://www.iso.org/standard/77845.html
- REF-OBC-08 - CharIN charging ecosystem and ISO 15118 resources: https://www.charin.global/technology/iso15118/
- REF-OBC-09 - Wolfspeed 25 kW high-efficiency bidirectional T-type converter: https://www.wolfspeed.com/products/power/reference-designs/crd-25bda6512n-k/

26. Appendix - Release Checklist

- Requirements review and architecture rationale approved.
- HWE.1/SWE.1 handoff and HSI timing/validity/owner defined.
- WCCA, DFMEA/DRBFM, safety/cyber analyses updated where applicable.
- All planned DVP cases have approved procedures and controlled equipment/configuration.
- All test failures have closed problem/deviation records and residual-risk disposition.
- RTM provides bidirectional link from source requirement through SYS.3/HWE.1/SWE.1 to verification and validation evidence.